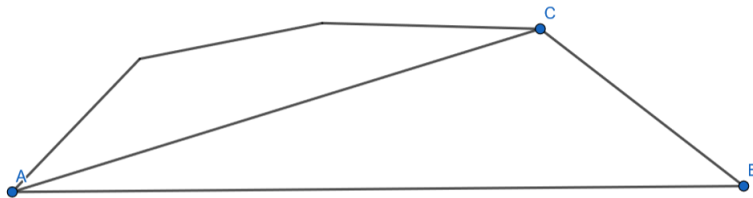


Geometric Optimization Article

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1 Introduction to Geometric Optimization

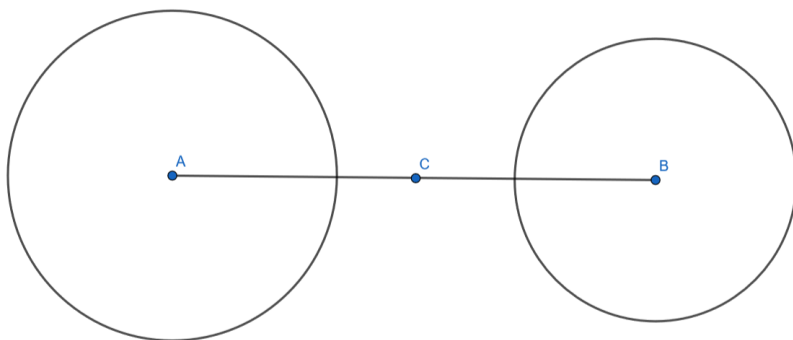
What is the shortest path between points A and B ? A straight line, right? But how do we prove that? Well, consider an alternative path consisting of a line or a series of lines, and take point C on that path. By the triangle inequality, $AB < AC + CB$, so thus AB must be the shortest path. $\square$, right?



But are we really finished? How do we know that the proof for the triangle inequality doesn't itself use the fact that the shortest distance between two points is a straight line? We must then first prove the triangle inequality.

1.1 Proof for the Triangle Inequality

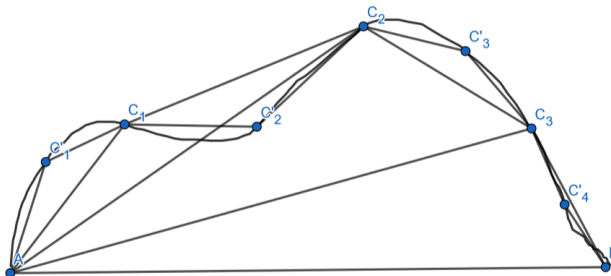
Suppose $\exists$ have three segments a, b, c such that $a + b \leq c$ and a, b, c are the sides of a triangle. Let us label the vertices opposite sides a, b, c to be A, B, C , respectively.



We can draw circles with radii a and b from B and A , respectively, and we notice that if $a + b < c$, then the circles don't intersect (since the distance between the closest two points will be $c - a - b$, so the closest points have distance between them), so no such point C can exist, and if $a + b = c$, then C lies on AB , implying $\triangle ABC$ does not exist.

1.2 Applying the Triangle Inequality

Now, we realize that if we have a series of straight lines that make up a possible alternate path between A and B , we can just apply the triangle inequality many times so that the straight line will be the shortest path.



And if we have a curve, we can choose some points (in the above case C_1, C_2 , and C_3) on the curve and prove that the path $AC_1C_2C_3B$ is longer than AB , only to choose some points C'_1, C'_2, C'_3, C'_4 on the curve as shown above and doing this process again. Doing this process infinitely will yield that the total path is shorter than AB .

As shown in the example above, what seems to just be a simple axiom is much more complicated, and hides a very interesting style of geometry not everyone learns in school or even college.

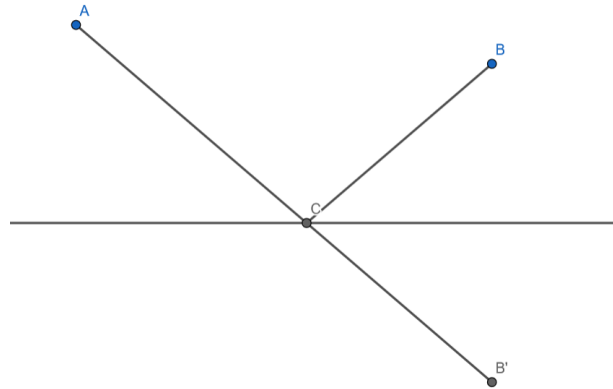
2 Transformations

2.1 Reflections

Suppose we have two points A and B , and a line l . What is the shortest path from A to l and then to B ?

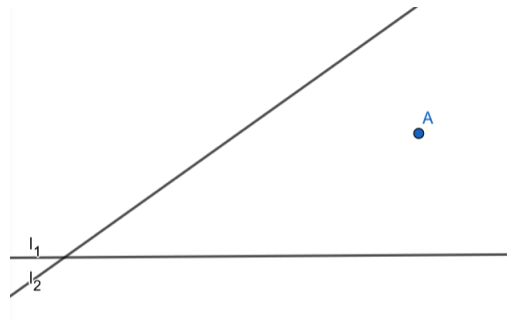


Well, we realize that if we touch l from A at point C , then if we reflect B over l to B' , $CB = CB'$, since l is the perpendicular bisector of BB' and C lies on l . Thus, we can derive our desired path from A to B using the optimal path from A to B' , which would just be a straight line by the Triangle Inequality.

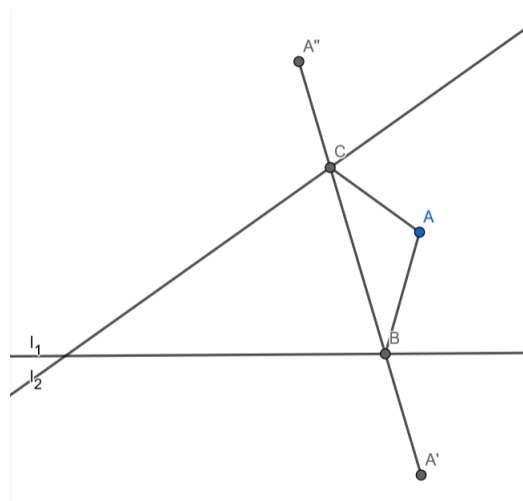


2.2 More Reflections

Now, suppose we have one point A and two intersecting lines l_1 and l_2 . What is the shortest path from A to l_1 , then to l_2 and back to A ?



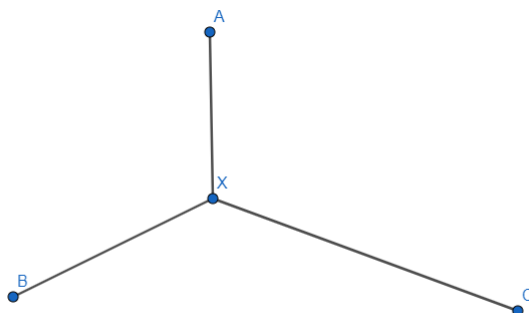
We can once again use reflections. We realize that if we reflect A over l_1 to A' , then if the supposed "optimal path" touches l_1 at B , then $AB = A'B$, and similarly for the point C on l_2 through which the optimal path will go through and A'' (the reflection of A over l_2). Thus, our points B and C are just the intersections of $A'A''$ with l_1 and l_2 , as shown below:



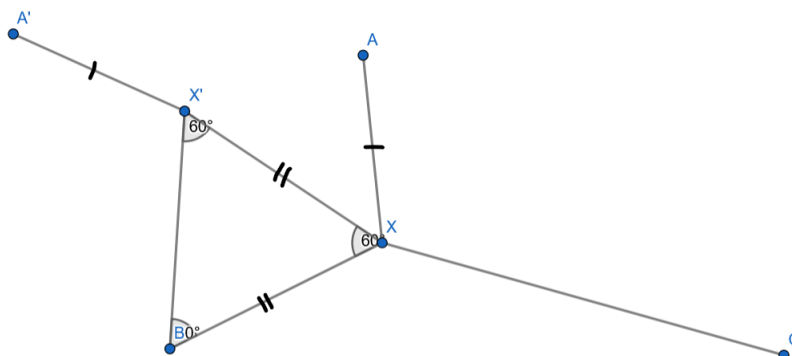
Now that we are more comfortable with reflections, we can tackle more difficult problems.

2.3 Rotations

Suppose you have 3 points A, B, C . What is the shortest path connecting them (for which point X is $AX + BX + CX$ minimized)?



Suppose we have point X inside $\triangle ABC$. Rotate $\overline{XA}$ 60° around B to segment $\overline{X'A'}$ counter-clockwise. We notice that since $\triangle BXX'$ is an equilateral triangle, $XB = XX'$. And since $\overline{X'A'}$ is just a rotation of $\overline{XA}$, $XA = X'A'$, so $CX + XX' + X'A' = AX + BX + CX$, our desired value. Since A' and C are fixed, the minimal distance $CX + XX' + X'A'$ is just a straight line from C to A' . Thus, $\angle AX'X = 120^\circ$, and since $\angle A'X'X$ is a rotation of $\angle AXB$, $\angle AXB = 120^\circ$. And since the reflection was symmetrical, we can do the rotation around C , and we find that X is the point for which $\angle AXB = \angle BXC = \angle CXA = 120^\circ$.



Now, how do we know such a point exists? Well, we can simply construct two circles with $\overline{AB}$ and $\overline{BC}$ as chords such that the major arcs of the chords are 240° . By the inscribed angle theorem, every point on these minor arcs will form a 120° angle, giving us our desired point.

One caveat, however, is that not all triangles have this point X inside $\triangle ABC$, namely those triangles with an angle larger than 120° . For these triangles, we can do the same thing, that is rotate $\overline{AX}$ 60° around B to $\overline{A'X'}$, after which we have that $CX + XX' + X'A'$ is equal to our desired value. (unfinished)

3 Area and Perimeter

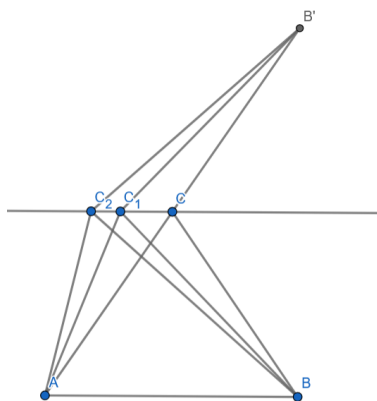
3.1 Basics

If we have a rectangle with a fixed area, how can we minimize the perimeter?

Let's say that the sides of the rectangle are a and b . We know that for perimeter p , $p^2 = (2a + 2b)^2 = 4(a + b)^2$. Following from that, we have that $p^2 - 16a = 4(a - b)^2 \implies p^2 = 16a + 4(a - b)^2 \implies p^2 \geq 16a$. This means that p^2 is minimized when $a = b$, which means that a square is optimal for minimizing perimeter.

3.2 Generalizing

How can we minimize the perimeter of triangles given a fixed area? If our base $(\overline{AB})$ and height are given, the length of AC and BC are minimized when $AC = BC$, since if we reflect B across the height line to B' , we find that $AC + BC = AC + B'C$ is minimized when A, C, B' are collinear, which makes $\triangle ABC$ isosceles (see diagram below).



We can conduct this process on all 3 sides, meaning that any pair of sides of $\triangle ABC$ is equal, so thus $\triangle ABC$ equilateral.

We can generalize this for all n sided polygons. We can take any two adjacent sides AB and BC , and we find that by the same logic as we used above, having $AB = BC$ minimized the perimeter of the polygon ABC , and since this will work for any adjacent pair of sides, the polygon is equilateral. However, proving that the polygon must be equiangular is a little harder.

If we say that each side has length x , we find that the area of all of our isosceles triangles is $\frac{x^2 \sin \theta}{2}$ for each angle θ between each of the pairs of sides, so we can factor out $\frac{x^2}{2}$ for each of the triangles, and find that the area is $\frac{x^2}{2} (\sin \theta_1 + \sin \theta_2 + \dots + \sin \theta_n)$. Now, if two of these angles aren't equal, say $\theta_i > \theta_j$ (we are assuming that all angles are less than 180°). Then, if we nudge θ_i down and nudge θ_j up, since $\sin' x = \cos x$, the total change in value would be $-\cos \theta_i + \cos \theta_j$ ($-\cos \theta_i$ since we are making θ_i smaller), which is positive since $\theta_i > \theta_j \implies \cos \theta_j > \cos \theta_i$. Thus, we maximize the area when the sines are closer together, or when all the angles are equal, so we have a regular polygon as the shape which minimizes perimeter for a fixed area.

4 Final Notes

Even if you have never heard the term before, you have most likely thought or heard about some of the general ideas covered in this article. Geometric optimization has lots of real world applications, such as in Molecular Modeling, where it is used to find the most stable shapes of molecules which minimizes the energy in the molecule, in computer graphics, where it is used to create the ideal mesh surface to create the most realistic 3D images, in design, where it is used to make components which satisfy certain criterion and conditions, minimize waste and are most efficient, and many

more. Even if you don't end up working with geometric optimization, it is a very interesting field, and even just in your head, you can visualize optimizing some shape or path (I can admit to doing this on the subway to Stuy).